

Phase 2: Elaboration — Phase Overview

Resilio — Mobile Mental Health Platform for Nepal | Milan Shrestha | ID: 23057135 | 02_Elaboration/README.md

RUP Phase: Elaboration | **Weeks:** 6–10 | **Dates:** December 2024

Phase Objective

Refine requirements through structured client engagement, produce the system architecture and all design artefacts, validate high-risk integrations through proof-of-concept prototypes (SuperTokens, eSewa, WebRTC), and deliver the Interim Report. Deliverables feed Chapter 2 (Background), Chapter 3 (Development), and the Appendices of the final report.

Documents in This Folder

File	Contents	Report Reference
`Client_and_Stakeholder_Documents.md`	Client profiles (Ms. Sushana Karki, Dr. Dibyandra Singh), interview guides, verbatim quotes, prototype review feedback, UAT summaries, requirements traceability matrix	Chapter 3, Section 3.2; Appendix F
`Database_Schema.md`	All 20 PostgreSQL tables — full column definitions, types, constraints, RLS policies, key indexes, 4-tier RLS summary	Chapter 3, Section 3.4; Appendix M
`API_Reference.md`	All REST endpoints with method, path, auth requirement, and JSON request/response examples; error format; auth middleware pattern	Chapter 3, Section 3.5; Appendix M

Key Artefacts Produced in Elaboration

Artefact	Status	Evidence Location
Interim Report	Done	`FYP Report Writing/Interim Report`
Software Requirements Specification (SRS)	Done	Appendix M
Use case diagrams (Customer, Therapist, Admin)	Done	Appendix J.5
Entity-Relationship Diagram	Done	Appendix J.8
System Architecture Diagram (3-tier)	Done	Appendix J.4
Activity diagrams (booking, video call, content)	Done	Appendix J.6
Sequence diagrams (OTP login, booking, WebRTC)	Done	Appendix J.7
Data Flow Diagrams (Level 0 + Level 1)	Done	Appendix J.9
Wireframes (7 screens)	Done	Appendix J.10
Figma high-fidelity mockups	Done	Appendix J.11
SuperTokens OTP prototype	Done	`Project Artefact/02 Requirement Analysis/Week - 08/`
WebRTC peer connection prototype	Done	`Project Artefact/02 Requirement Analysis/Week - 09/`
eSewa payment prototype	Done	`Project Artefact/02 Requirement Analysis/Week - 08/`
Database schema (all 20 tables)	Done	`Database_Schema.md` (this folder)

Key Decisions Made

Decision	Rationale
Feature-first Clean Architecture	Enables parallel development of Customer/Therapist/Admin modules with clear domain boundaries

Decision	Rationale
BLoC/Cubit state management	Predictable unidirectional data flow; mature Flutter ecosystem support
PostgreSQL RLS (4-tier policy)	Enforce data isolation at the database level, not just application level — clinical data requires this
Cloudinary for media	Free tier, CDN delivery, Dart SDK available; avoids Supabase storage bandwidth limits
SQLite + Protobuf for offline	Binary serialisation (vs JSON) reduces SQLite storage size for offline mood entries